

Electrónica II

Dispositivos Comerciales



Diodos de potencia



PRIMARY CHARACTERISTICS

$I_{F(AV)}$	3.0 A
V_{RRM}	20 V, 30 V, 40 V
I_{FSM}	80 A
V_F	0.475 V, 0.500 V, 0.525 V
$T_J \text{ max.}$	125 °C
Package	DO-201AD
Diode variations	Single



www.vishay.com

1N5820, 1N5821, 1N5822

Vishay General Semiconductor

Schottky Barrier Plastic Rectifier



DO-201AD

FEATURES

- Guarding for overvoltage protection
- Very small conduction losses
- Extremely fast switching
- Low forward voltage drop
- High frequency operation
- Solder dip 275 °C max. 10 s, per JESD 22-B106
- Material categorization: For definitions of compliance please see www.vishay.com/doc/799312



RoHS
compliant

TYPICAL APPLICATIONS

For use in low voltage high frequency inverters, freewheeling, DC/DC converters, and polarity protection applications.

MECHANICAL DATA

Case: DO-201AD
Molding compound meets UL 94 V-0 flammability rating
Base P/N-E3 - RoHS-compliant, commercial grade
Terminals: Matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
E3 suffix meets JESD 201 class 1A whisker test
Polarity: Color band denotes the cathode end

MAXIMUM RATINGS ($T_A = 25$ °C unless otherwise noted)

PARAMETER	SYMBOL	1N5820	1N5821	1N5822	UNIT
Maximum repetitive peak reverse voltage	V_{RRM}	20	30	40	V
Maximum RMS voltage	V_{RRM}	14	21	28	V
Maximum DC blocking voltage	V_{DC}	20	30	40	V
Non-repetitive peak reverse voltage	V_{RRM}	24	36	48	V
Maximum average forward rectified current at 0.375" (9.5 mm) lead length at $T_J = 95$ °C	$I_{F(AV)}$	3.0			A
Peak forward surge current, 8.3 ms single half sine-wave superimposed on rated load	I_{FSM}	80			A
Operating junction and storage temperature range	T_J, T_{STG}	- 65 to +125			°C

ELECTRICAL CHARACTERISTICS ($T_A = 25$ °C unless otherwise noted)

PARAMETER	TEST CONDITIONS	SYMBOL	1N5820	1N5821	1N5822	UNIT
Maximum instantaneous forward voltage	3.0	$V_F^{(1)}$	0.475	0.500	0.525	V
Maximum instantaneous forward voltage	9.4	$V_F^{(1)}$	0.850	0.900	0.950	V
Maximum average reverse current at rated DC blocking voltage	$T_A = 25$ °C $T_A = 100$ °C	$I_R^{(1)}$	2.0 20			mA

Note

⁽¹⁾ Pulse test: 300 μ s pulse width, 1 % duty cycle

Diodos de potencia



PRIMARY CHARACTERISTICS

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I_{FSM}	80 A
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Note

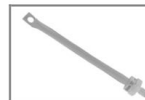
⁽¹⁾ Pulse test: 300 μ s pulse width, 1 % duty cycle

Diodos de potencia



V_{RSM} V	V_{RRM} V	$I_{FRMS} = 80$ A (maximum value for continuous operation) $I_{FAV} = 45$ A (sin. 180; $T_C = 125$ °C)	
400	400	SKN 45/04	SKR 45/04
800	800	SKN 45/08	SKR 45/08
1200	1200	SKN 45/12	SKR 45/12
1400	1400	SKN 45/14	SKR 45/14
1600	1600	SKN 45/16	SKR 45/16

SKN 45, SKR 45



Stud Diode

V_{RSM} V	V_{RRM} V	$I_{FRMS} = 80$ A (maximum value for continuous operation) $I_{FAV} = 45$ A (sin. 180; $T_C = 125$ °C)	
400	400	SKN 45/04	SKR 45/04
800	800	SKN 45/08	SKR 45/08
1200	1200	SKN 45/12	SKR 45/12
1400	1400	SKN 45/14	SKR 45/14
1600	1600	SKN 45/16	SKR 45/16

Rectifier Diode

SKN 45
SKR 45

Features

- Reverse voltages up to 1600 V
- Hermetic metal case with glass insulator
- Cooling via heatsinks
- Threaded stud ISO M8 or 1/4 - 28 UNF 2A
- SKN:** anode to stud
- SKR:** cathode to stud

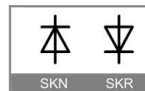
Typical Applications *

- All purpose high power rectifier diode
- Non-controllable and half-controllable rectifiers
- Free-wheeling diodes
- Recommended snubber network:

R_C : 0.1 μ F, 100 Ω ($P_R = 1$ W),
 R_L : 80 k Ω ($P_R = 6$ W)

1) Mounting with grease-like thermal compound or joint contact compound

Symbol	Condition	Values	Units
I_{FAV}	sin. 180; $T_C = 117$ °C	50	A
I_D	K 5; $T_A = 45$ °C; B2 / B6	40 / 57	A
	K 1,1; $T_A = 45$ °C; B2 / B6	86 / 120	A
I_{FSM}	$T_A = 25$ °C; 10 ms	700	A
	$T_A = 180$ °C; 10 ms	600	A
I_T	$T_A = 25$ °C; 8.3...10 ms	2500	A ² s
	$T_A = 180$ °C; 8.3...10 ms	1800	A ² s
V_F	$T_A = 25$ °C; $I_F = 150$ A	max. 1.6	V
$V_{(T_C)}$	$T_A = 180$ °C	max. 0.85	V
r_i	$T_A = 180$ °C	max. 5	m Ω
I_{HD}	$T_A = 180$ °C; $V_{HD} = V_{RRM}$	max. 10	mA
Q_{rr}	$T_A = 160$ °C; $-di/dt = 10$ A/ μ s	70	μ C
$R_{rec(1)}$		0.85	KW
$R_{rec(2)}$		0.25	KW
T_{stg}	-40...+180 °C		°C
	-55...+180 °C		°C
V_{M8}	M8 Stud	-	V-
M_8	1/4 - 28 UNF 2A	4	Nm
	M8 Stud (lubricated) ¹⁾	2.5	Nm
	1/4 - 28 UNF 2A (lubricated) ¹⁾	3	Nm
a		2	Nm
m	approx.	5 * 9.81	m/s ²
		30	g
Case		E 12	



Diodos de potencia



Diodos de potencia



VOLTAGE RATINGS

Part Number	$V_{RRM} + V_R$ (V) Max. rep. peak reverse voltage		$V_{RSM} + V_R$ (V) Max. non- rep. peak reverse voltage
	$T_J = 0 \text{ to } 180^\circ\text{C}$	$T_J = -40 \text{ to } 0^\circ\text{C}$	$T_J = 25 \text{ to } 180^\circ\text{C}$
A1A:260.02	200	200	300
A1A:260.04	400	400	500
A1A:260.06	600	600	700
A1A:260.08	800	800	900
A1A:260.10	1000	1000	1100
A1A:260.12	1200	1200	1300
A1A:260.14	1400	1400	1500
A1A:260.16	1600	1600	1700



A1A:260.XX

VOLTAGE RATINGS

Part Number	$V_{RRM} + V_R$ (V) Max. rep. peak reverse voltage		$V_{RSM} + V_R$ (V) Max. non- rep. peak reverse voltage
	$T_J = 0 \text{ to } 180^\circ\text{C}$	$T_J = -40 \text{ to } 0^\circ\text{C}$	$T_J = 25 \text{ to } 180^\circ\text{C}$
A1A:260.02	200	200	300
A1A:260.04	400	400	500
A1A:260.06	600	600	700
A1A:260.08	800	800	900
A1A:260.10	1000	1000	1100
A1A:260.12	1200	1200	1300
A1A:260.14	1400	1400	1500
A1A:260.16	1600	1600	1700

This datasheet applies to:

Metric thread: A1A:260.XX,
A1B:260.XX

Inch thread: A2A:260.XX,
A2B:260.XX

MAXIMUM ALLOWABLE RATINGS

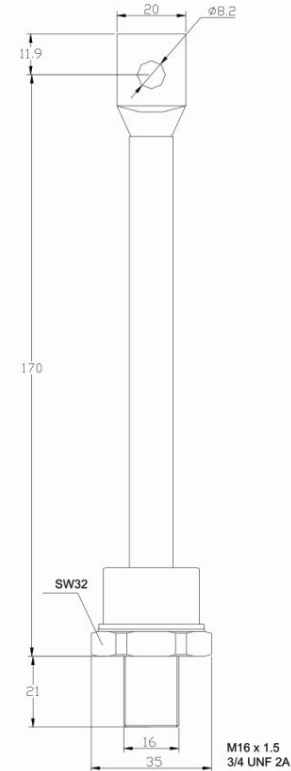
PARAMETER	VALUE	UNITS	NOTES
T_J Junction Temperature	-40 to 180	$^\circ\text{C}$	-
T_{stg} Storage Temperature	-40 to 180	$^\circ\text{C}$	-
I_{AV} Max. Av. current	260	A	180° half sine wave
I_{AV} @ Max. T_J	125	$^\circ\text{C}$	-
I_{RMS} Norm. RMS current	530	A	-
I_{FSM} Max. Peak non- rep. surge current	5048	A	50 Hz half cycle sine wave Initial $T_J = 180^\circ\text{C}$, rated V_{RRM} applied after surge.
	5500		60 Hz half cycle sine wave Initial $T_J = 180^\circ\text{C}$, no voltage applied after surge.
	6000		50 Hz half cycle sine wave Initial $T_J = 180^\circ\text{C}$, no voltage applied after surge.
	6540		60 Hz half cycle sine wave Initial $T_J = 180^\circ\text{C}$, no voltage applied after surge.
t^1 Max. t^1 capability	116	ms	Initial $T_J = 180^\circ\text{C}$, rated V_{RRM} applied after surge.
	126		$t = 8.3 \text{ ms}$
	163		$t = 10 \text{ ms}$ Initial $T_J = 180^\circ\text{C}$, no voltage applied after surge.
	178		$t = 8.3 \text{ ms}$
t^2 Max. t^2 capability	1350	ms^2	Initial $T_J = 180^\circ\text{C}$, no voltage applied after surge.
F Mounting Force	30(-287)	N,mq(lb,in)	t^2 for time $t_s = t^2 t_s^{-1/2}$ ($t_s < 10 \text{ ms}$)

Diodos de potencia



VOLTAGE RATINGS

Part Number	$V_{RRM} \cdot V_R$ (V) Max. rep. peak reverse voltage		$V_{RSM} \cdot V_R$ (V) Max. non- rep. peak reverse voltage
	$T_J = 0$ to 180°C	$T_J = -40$ to 0°C	
	$T_J = 0$ to 180°C	$T_J = -40$ to 0°C	$T_J = 25$ to 180°C
A1A:260.02	200	200	300
A1A:260.04	400	400	500
A1A:260.06	600	600	700
A1A:260.08	800	800	900
A1A:260.10	1000	1000	1100
A1A:260.12	1200	1200	1300
A1A:260.14	1400	1400	1500
A1A:260.16	1600	1600	1700



Tiristores de potencia



V_{RSM} V	V_{RRM}, V_{DRM} V	$I_{TRMS} = 450 \text{ A}$ (maximum value for continuous operation) $I_{TAV} = 250 \text{ A}$ (sin. 180; $T_c = 85^\circ \text{C}$)	
500	400	SKT 250/04D	
900	800	SKT 250/08D	
1300	1200	SKT 250/12E	
1500	1400	SKT 250/14E	
1700	1600	SKT 250/16E	

SKT 250



Stud Thyristor

Line Thyristor

SKT 250

Features

- Hermetic metal case with glass insulator
- Threaded stud ISO M24x1,5
- High R and I_{TSM} values for easy fusing
- International standard case

Typical Applications*

- DC motor control (e. g. for machine tools)
- Controlled rectifiers (e. g. for battery charging)
- AC controllers (e. g. for temperature control)
- Recommended snubber network
e. g. for $V_{RRM} \leq 400 \text{ V}$:
 $R = 33 \Omega / 32 \text{ W}$, $C = 0.47 \mu\text{F}$

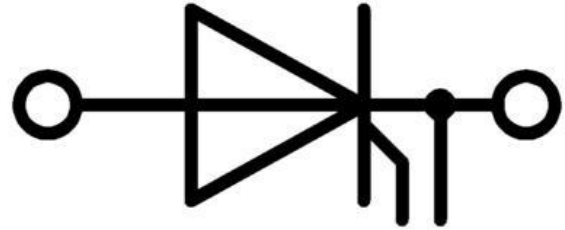
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900	800	SKT 250/08D	
1300	1200	SKT 250/12E	
1500	1400	SKT 250/14E	
1700	1600	SKT 250/16E	

Symbol	Conditions	Values	Units
I_{TAV}	sin. 180; $T_c = 100$ (85) $^\circ \text{C}$; K0.55; $T_c = 45^\circ \text{C}$; B2 / B6 K0.55; $T_c = 35^\circ \text{C}$; B2 / B6	185 (250) 240 / 330 490 / 675	A
I_{TSM}	K0.55; $T_c = 45^\circ \text{C}$; W1C	265	A
I_{TSM}	$T_c = 25^\circ \text{C}$; 10 ms $T_c = 130^\circ \text{C}$; 10 ms $T_c = 25^\circ \text{C}$; 8.35 ... 10 ms $T_c = 130^\circ \text{C}$; 8.35 ... 10 ms	7000 6000 245000 180000	A
V_T	$T_c = 25^\circ \text{C}$; $I_T = 800 \text{ A}$	max. 1.65	V
$V_{T(110)}$	$T_c = 130^\circ \text{C}$	max. 1	V
r_T	$T_c = 130^\circ \text{C}$	max. 0.7	m Ω
$t_{T2} - t_{T0}$	$T_c = 130^\circ \text{C}$; $V_{DRM} = V_{RRM}$; $V_{DRM} = V_{DRM}$	max. 50	mA
t_{T2}	$T_c = 25^\circ \text{C}$; $I_T = 1 \text{ A}$; $dI_T/dt = 1 \text{ A}/\mu\text{s}$	1	μs
t_T	$V_D = 0.67 \cdot V_{DRM}$	2	μs
$(dI/dt)_{T2}$	$T_c = 130^\circ \text{C}$	max. 100	A/ μs
$(dI/dt)_{T0}$	$T_c = 130^\circ \text{C}$; SKT...D / SKT...E	max. 500 / 1000	A/ μs
t_{tr}	$T_c = 130^\circ \text{C}$	50 ... 150	μs
t_{tr}	$T_c = 25^\circ \text{C}$; typ. / max.	150 / 250	mA
t_{tr}	$T_c = 25^\circ \text{C}$; $R_{th} = 33 \Omega$; typ. / max.	300 / 600	mA
V_{GT}	$T_c = 25^\circ \text{C}$; d.c.	min. 3	V
V_{GT}	$T_c = 25^\circ \text{C}$; d.c.	min. 200	V
V_{DO}	$T_c = 130^\circ \text{C}$; d.c.	max. 0.25	V
t_{off}	$T_c = 130^\circ \text{C}$; d.c.	max. 10	mA
$R_{th(j-a)}$	corr.	0.11	K/W
$R_{th(j-a)}$	sin. 180	0.123	K/W
$R_{th(j-a)}$	rec. 120	0.137	K/W
$R_{th(j-a)}$		0.015	K/W
T_{st}		-40 ... +130	$^\circ \text{C}$
T_{st}		-55 ... +150	$^\circ \text{C}$
V_{max}		60	Nm
M_{max}	to heatink	5 * 9.81	mm ²
m	approx.	490	g
Case		B 7	

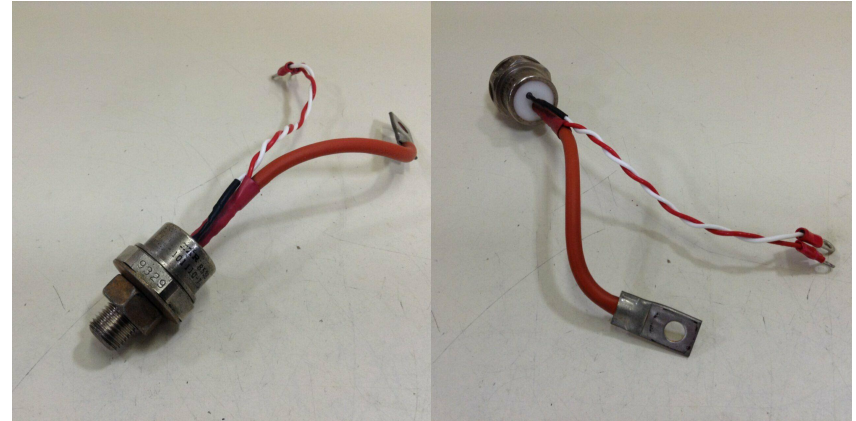


SKT

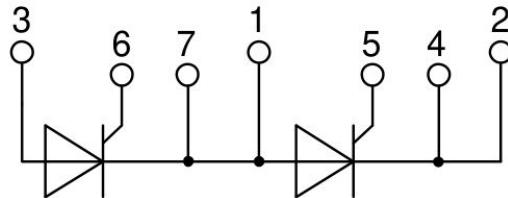
Tiristores de potencia



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1700	1600	SKT 250/16E		



Tiristores de potencia



Thyristor Module

Phase leg

Part number
MCC162-16io1

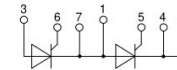
MCC162-16io1

$V_{RRM} = 2 \times 1600 \text{ V}$
 $I_{TAV} = 181 \text{ A}$
 $V_T = 1.03 \text{ V}$



Backside: isolated

E72873



Features / Advantages:

- Thyristor for line frequency
- Planar passivated chip
- Long term stability
- Direct Copper Bonded Al2O3-ceramic

Applications:

- Line rectifying 50/60 Hz
- Softstart AC motor control
- DC Motor control
- Power converter
- AC power control
- Lighting and temperature control

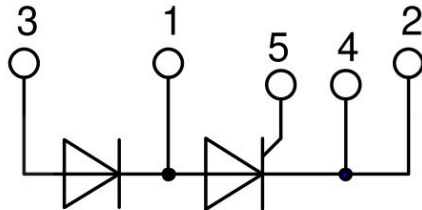
Package: Y4

- Isolation Voltage: 3600 V~
- Industry standard outline
- RoHS compliant
- Soldering pins for PCB mounting
- Base plate: DCB ceramic
- Reduced weight
- Advanced power cycling

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Tiristores de potencia



MCD162-16Io1

Thyristor \ Diode Module

$V_{RRM} = 2 \times 1600 \text{ V}$
 $I_{TAV} = 181 \text{ A}$
 $V_T = 1.03 \text{ V}$

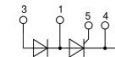
Phase leg

Part number
MCD162-16Io1



Backside: isolated

IXYS E72873



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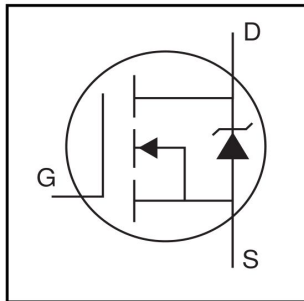
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MOSFET de potencia



$$V_{DSS} = 200V$$

$$R_{DS(on)} = 0.075\Omega$$

$$I_D = 30A$$

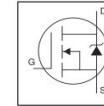
International
IR Rectifier

PD - 94008

IRFP250N

HEXFET® Power MOSFET

- Advanced Process Technology
- Dynamic dv/dt Rating
- 175°C Operating Temperature
- Fast Switching
- Fully Avalanche Rated
- Ease of Paralleling
- Simple Drive Requirements



$$V_{DSS} = 200V$$

$$R_{DS(on)} = 0.075\Omega$$

$$I_D = 30A$$

Description

Fifth-Generation HEXFETs from International Rectifier utilize advanced processing techniques to achieve extremely low on-resistance per silicon area. This benefit, combined with the fast switching speed and ruggedized device design that HEXFET Power MOSFETs are well known for, provides the designer with an extremely efficient and reliable device for use in a wide variety of applications.

The TO-247 package is preferred for commercial-industrial applications where higher power levels preclude the use of TO-220 devices. The TO-247 is similar but superior to the earlier TO-218 package because of its isolated mounting hole.



TO-247AC

Absolute Maximum Ratings

Parameter	Max.	Units
I_D @ $T_C = 25^\circ C$	Continuous Drain Current, $V_{GS} @ 10V$	30
I_D @ $T_C = 100^\circ C$	Continuous Drain Current, $V_{GS} @ 10V$	21
I_{DM}	Pulsed Drain Current (1)	120
P_D @ $T_C = 25^\circ C$	Power Dissipation	214
	Linear Derating Factor	1.4
V_{GS}	Gate-to-Source Voltage	± 20
E_{AS}	Single Pulse Avalanche Energy (2)	315
I_{AS}	Avalanche Current (2)	30
E_{AR}	Repetitive Avalanche Energy (2)	21
dv/dt	Peak Diode Recovery dv/dt (3)	8.6
T_J	Operating Junction and Storage Temperature Range	-55 to $+175$
T_{SOL}	Soldering Temperature, for 10 seconds	300 (1.6mm from case)
	Mounting Torque, 6-32 or M3 screw	18 lbf-in (1.19Nm)

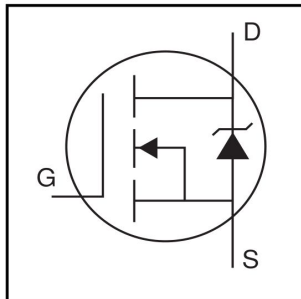
Thermal Resistance

Parameter	Typ.	Max.	Units
R_{JC}	Junction-to-Case	0.7	$^\circ C/W$
R_{CS}	Case-to-Sink, Flat, Greased Surface	0.24	$^\circ C/W$
R_{JA}	Junction-to-Ambient	—	40

www.irf.com

10/09/00

MOSFET de potencia



$$V_{DSS} = 500V$$

$$R_{DS(on)} = 0.48 \Omega$$

$$I_D = 13A$$

Truesemi

TSP13N50M/TSF13N50M
500V N-Channel MOSFET

General Description

This Power MOSFET is produced using Truesemi's advanced planar stripe DMOS technology. This advanced technology has been especially tailored to minimize on-state resistance, provide superior switching performance, and withstand high energy pulse in the avalanche and commutation mode. These devices are well suited for high efficiency switched mode power supplies, active power factor correction based on half bridge topology.

Features

- 13A, 500V, Max. $R_{DS(on)} = 0.48 \Omega$ @ $V_{GS} = 10V$
- Low gate charge (typical 45nC)
- High ruggedness
- Fast switching
- 100% avalanche tested
- Improved dv/dt capability

Absolute Maximum Ratings $T_J = 25^\circ C$ unless otherwise specified

Symbol	Parameter	TSP13N50M	TSF13N50M	Units
V_{DS}	Drain-Source Voltage	500		V
V_{GS}	Gate-Source Voltage		± 30	V
I_D	Drain Current	13	13*	A
		$T_J = 100^\circ C$	8	8*
I_{DS}	Pulsed Drain Current	52	52*	A
E_{AS}	Single Pulsed Avalanche Energy	860		mJ
E_{AR}	Repetitive Avalanche Energy	19.5		mJ
dv/dt	Peak Diode Recovery dv/dt	4.5		V/ns
P_D	Power Dissipation ($T_J = 25^\circ C$)	195	48	W
	-Dense above $25^\circ C$	1.56	0.39	W/ $^\circ C$
T_J, T_{STG}	Operating and Storage Temperature Range		-55 to +150	$^\circ C$
T_L	Maximum lead temperature for soldering purposes, 1/8" from case for 5 seconds	300		$^\circ C$

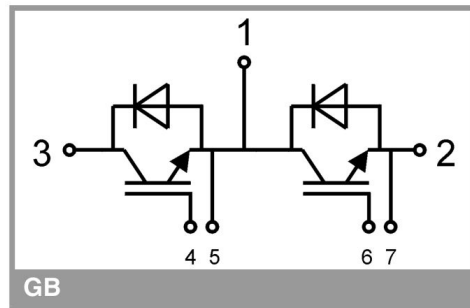
* Drain current limited by maximum junction temperature.

Thermal Resistance Characteristics

Symbol	Parameter	TSP13N50M	TSF13N50M	Units
R_{JC}	Thermal Resistance, Junction-to-Case	0.64	2.58	$^\circ C/W$
R_{CS}	Thermal Resistance, Case-to-Sink Typ.	0.5	---	$^\circ C/W$
R_{JA}	Thermal Resistance, Junction-to-Ambient	62.5	62.5	$^\circ C/W$

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IGBT



SKM100GB12T4



SEMITRANS® 2

Fast IGBT Modules

SKM100GB12T4

Features

- IGBT4 = 4. generation fast trench IGBT (Infineon)
- CAL4 = Soft switching 4. generation CAL-diode
- Isolated copper baseplate using DBC technology (Direct Bonded Copper)
- Increased power cycling capability
- With integrated gate resistor
- For higher switching frequencies up to 20kHz
- UL recognized, file no. E63532

Typical Applications*

- AC inverter drives
- UPS
- Electronic welders at low up to 20 kHz

Remarks

- Case temperature limited to $T_c = 125^\circ\text{C}$ max.
- Recommended $T_{\text{amb}} = -40 \dots +150^\circ\text{C}$
- Product reliability results valid for $T_c = 150^\circ\text{C}$

Absolute Maximum Ratings

Symbol	Conditions	Values	Unit
IGBT			
V_{CES}	$T_c = 25^\circ\text{C}$	1200	V
I_c	$T_c = 175^\circ\text{C}$	160	A
	$T_c = 80^\circ\text{C}$	123	A
I_{CRM}		100	A
I_{CRM}	$I_{\text{CRM}} = 3M_{\text{CRM}}$	300	A
V_{CES}		-20 ... 20	V
t_{spe}	$V_{\text{GE}} = 800\text{ V}$	10	μs
	$V_{\text{GE}} \leq 15\text{ V}$ $V_{\text{CES}} \leq 1200\text{ V}$	$T_c = 150^\circ\text{C}$	
T_j		-40 ... 175	$^\circ\text{C}$

Inverse diode

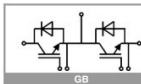
I_F	$T_c = 175^\circ\text{C}$	$T_c = 25^\circ\text{C}$	121	A
	$T_c = 80^\circ\text{C}$		91	A
I_{FRM}			100	A
I_{FRM}	$I_{\text{FRM}} = 2M_{\text{FRM}}$		300	A
t_{RR}	$t_{\text{RR}} = 10\text{ ms}$, $\sin 180^\circ$, $T_c = 25^\circ\text{C}$		550	A
T_j			-40 ... 175	$^\circ\text{C}$

Module

T_{terminal}	$T_{\text{terminal}} = 80^\circ\text{C}$	200	A
T_{sp}		-40 ... 125	$^\circ\text{C}$
V_{test}	AC sinus 50 Hz, $t = 1\text{ min}$	4000	V

Characteristics

Symbol	Conditions	min.	typ.	max.	Unit
IGBT					
$V_{\text{CE(sat)}}$	$I_c = 100\text{ A}$		1.80	2.05	V
	$V_{\text{GE}} = 15\text{ V}$ chiplevel		2.20	2.40	V
V_{CE}	$T_c = 25^\circ\text{C}$		0.8	0.9	V
	chiplevel		0.7	0.8	V
r_{CE}	$V_{\text{GE}} = 15\text{ V}$		10.00	11.50	m Ω
	chiplevel		15.00	16.00	m Ω
$V_{\text{GE(sat)}}$	$V_{\text{GE}} = V_{\text{CE}}$, $I_c = 3.8\text{ mA}$	5	5.8	6.5	V
	$V_{\text{GE}} = 0\text{ V}$				
I_{CES}	$T_c = 25^\circ\text{C}$			1	mA
	$V_{\text{CE}} = 1200\text{ V}$				
C_{iss}	$V_{\text{GE}} = 25\text{ V}$		6.15		nF
	$V_{\text{GE}} = 0\text{ V}$		0.40		nF
C_{oss}	$f = 1\text{ MHz}$		0.345		nF
	$V_{\text{GE}} = -8\text{ V} \dots +15\text{ V}$		565		pC
Q_{isr}			7.5		(Ω)
t_{turn}	$V_{\text{CE}} = 600\text{ V}$		165		ns
	$I_c = 100\text{ A}$				
t_c	$V_{\text{GE}} = +15\text{ V}$	$T_c = 150^\circ\text{C}$	47		ns
	$T_c = 150^\circ\text{C}$		15		ns
E_{on}	$R_{\text{DS(on)}} = 1.1\text{ }\Omega$	$T_c = 150^\circ\text{C}$	400		ns
t_{off}		$T_c = 150^\circ\text{C}$	75		ns
t_{d}	$dI/dt_{\text{on}} = 1800\text{ A}/\mu\text{s}$	$T_c = 150^\circ\text{C}$	10.2		mJ
E_{off}	$dI/dt_{\text{off}} = 1130\text{ A}/\mu\text{s}$	$T_c = 150^\circ\text{C}$			mJ
$R_{\text{th(j-a)}}$	per IGBT		0.27		K/W



Referencias

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- SEMIKRON - <https://www.semikron.com/>
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- IXYS - <https://www.ixys.com/>
- AEGIS Semiconductores LTDA

